

Advanced Glycation End Products in Foods and a Practical Guide to Their Reduction in the Diet

Modern diets are largely heat-processed and as a result contain high levels of advanced glycation end products (AGEs). Dietary advanced glycation end products (dAGEs) are known to contribute to increased oxidant stress and inflammation, which are linked to the recent epidemics of diabetes and cardiovascular disease. This report significantly expands the available dAGE database, validates the dAGE testing methodology, compares cooking procedures and inhibitory agents on new dAGE formation, and introduces practical approaches for reducing dAGE consumption in daily life. Based on the findings, dry heat promotes new dAGE formation by >10- to 100-fold above the uncooked state across food categories. Animal-derived foods that are high in fat and protein are generally AGE-rich and prone to new AGE formation during cooking. In contrast, carbohydrate-rich foods such as vegetables, fruits, whole grains, and milk contain relatively few AGEs, even after cooking. The formation of new dAGEs during cooking was prevented by the AGE inhibitory compound aminoguanidine and significantly reduced by cooking with moist heat, using shorter cooking times, cooking at lower temperatures, and by use of acidic ingredients such as lemon juice or vinegar. The new dAGE database provides a valuable instrument for estimating dAGE intake and for guiding food choices to reduce dAGE intake.